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Hedging Currency Risk at TT Textiles

It was a hot March morning in Kolkata in the year 2009. Sanjay K. Jain, —Joint Managing Director of TT Textiles, watched the sunlight stream in through his office windowpane. But his mind was elsewhere, tracking the movements of the Swiss franc (CHF) in the last few months and the world events that had caused them. The Swiss franc had touched 1.17 CHF/US$ from the previous year's record of 0.96CHF/US$. That was good news for him. Or was it? The irony of the situation was not lost on him. Once, the Swiss had franc barely figured among all the different currencies that vied for his attention in the normal course of things. Yet, lately, it was the movement of the CHF that weighed on his mind most heavily.

As an exporter to more than 30 countries, TT Textiles was no newcomer to the area of currency risk. TT Textiles usually used forwards to manage currency risks. However, during 2006-07, when the INR was expected to appreciate to an unprecedented high of 35 INR/US$, the company had entered into a swap deal based on the historical stability of the CHF against the US$. At the time, the deal had looked relatively safe and very lucrative. However, when the global financial crisis struck in 2008, it started making sizeable mark-to-market losses. Luckily it turned around in 2009 and was no longer in the red. But with three months left on the contract, the big question Jain faced was whether to quit now or hold it till maturity.

BACKGROUND

The Textile Industry

The textile and clothing industry in India had traditionally been an export-oriented industry. In 2008, it contributed four per cent to the overall GDP of India and accounted for 14 per cent of the industrial production and 14 per cent of total exports of goods[1]. More importantly, India earned about 27 per cent of its total foreign exchange through textile exports. It was also the second largest employer after agriculture, providing direct employment to 35 million people and indirect employment to 45 million people. In 2008-09, the total sales generated by the textile and clothing sector amounted to US$33.4 billion from the domestic market and US$21.6 billion from exports.

[1] http://texmin.nic.in/annualrep/ar_08_09_english.pdf *last accessed on February 14, 2013.*

Professor Rajesh Chakrabarti prepared this case solely as a basis for class discussion. This case is not intended to serve as an endorsement, a source of primary data, or an illustration of effective or ineffective management. The author thanks Shashvat Rai and Anurag Sharma from ISB's PGP Class of 2011 for assisting in the writing of this case. This case was developed under the aegis of the Centre for Teaching, Learning, and Case Development, ISB.

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As a mature industry, the textile industry was marked by relatively low margins — varying from three per cent to 12 per cent depending on where in the value chain a specific company operated.

The total market for textiles and clothing was expected to reach US$100 billion by 2015, with 43 per cent of revenues coming from exports. Specifically, textile exports were expected to yield US$22 billion and the domestic textile market was expected to yield US$28 billion by 2015.[2]

The US dollar was the dominant currency for pricing textile products worldwide, in a large measure even for exports to Europe or Latin American countries.

TT Textiles Ltd

TT Textiles Limited, the flagship company of the TT Group, was founded in 1978 by the family of Dr. Rikhab Chand Jain. It was India's first knitwear company to go public. TT Textiles was a vertically integrated textile company with a presence in the entire cotton chain, from fibre to yarn to knitted fabric and garments. It had manufacturing facilities in all the major garment centers —Tirupur, Kolkata, Delhi, Varanasi, Saharanpur and Kanpur. It had ginning units in Gondal, Gujarat and branches for cotton in Jalna, Maharashtra. The company's core businesses were agrocommodity, cotton, yarn, fabric and garments, and its markets were spread all over the world, as shown below (also see Exhibit 1):

- **Raw Cotton:** India, Bangladesh, Pakistan, China, Korea, Taiwan, Vietnam, Thailand, Indonesia, Malaysia, Turkey and Hong Kong.
- **Yarn:** India, Korea, Taiwan, China, Hong Kong, Malaysia, Indonesia, Singapore, Bangladesh, Mauritius, Egypt, Turkey, Israel, Italy, Colombia, Vietnam, Brazil, USA, Peru, Argentina, Slovenia, Spain, Portugal, Italy, Germany, South Africa, Honduras, Guatemala, Tunisia, Morocco and Tanzania.
- **Fabric:** India, Bangladesh, Europe and USA.
- **Inner and Casual Wear (Garments):** India, USA, Europe and the Middle East.
- **Agro-commodity:** Vietnam, Serbia, Malaysia, Bangladesh, Korea, India and Turkey.

Sanjay Jain, an MBA gold medalist from IIM, Ahmedabad and an Associate Member of the Institute of Company Secretaries of India (ACS) and Institute of Cost Accountants of India (AICWA), began his career at ICICI Bank before starting his own brokerage firm, which he later sold. He joined TT Textiles in 2001 and was instrumental in expanding the textile business and setting up the marketing network for raw cotton yarn in over 20 countries around the world.

Approximately 75 per cent of TT Textiles' revenues came from exports, and at any particular point of time, the company had an exposure of roughly US$25 million. The life of a typical export transaction in the industry — particularly of the kind that TT was party to — was less than three months. TT Textiles enjoyed a margin of five to six per cent in its business.

RISE OF CURRENCY DERIVATIVE PRODUCTS IN INDIA

Currency derivative products were relatively new entrants in India. Most Indian companies depended on their banks to hedge currency exposures. In a 2009 newspaper article, Ramesh Kumar, Senior Vice President and Head, Debt and Currency Markets of Asit C. Mehta, explained:

[2] *Implicit in the figures above is an assumption of a CAGR of eight per cent for textile exports and 10 per cent for textile domestic demand.*

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> Historically, in a controlled environment, India Inc. relied on banks for covering its foreign exchange requirements. … Some of the companies trade actively in foreign exchange and have a separate treasury management unit for foreign exchange transactions. However, there are also large numbers of small and medium enterprises which participate in the currency market passively and depend on commercial banks (authorised dealers) for their requirement of foreign exchange and coverage of currency exposure.[3]

Exchange-traded Currency Derivatives

The currency market was one of India's biggest financial markets, with turnover on the spot and forward markets together yielding around US$12 billion a day in April 2007. Since September 2008, there had been both foreign exchange (forex) forwards as well as futures markets in the country trading the INR-US$. Derivatives on other currencies were not traded. The rupee-dollar forward market was an over-the-counter (OTC) market, the trades on which were settled through the Clearing Corporation of India Ltd (CCIL), which was the clearing house for forex and interest rate trades in India. This minimized the credit risk associated with these agreements in the Indian market. According to Chakrabarti and De, "In 2006-07, 85,106 forex forward transactions went to CCIL for settlement, with a notional value of US$342 billion. By late 2006, forward market turnover was nudging US$2 billion a day. Foreign institutional investors were able to do transactions on the currency derivatives market that could be characterized as 'hedging' of the currency risk exposure on their Indian investment."[4]

In addition to the domestic rupee-dollar forward market, there was active trading for cash-settled rupee-dollar forwards in Hong Kong, Singapore, Dubai and London on what were termed "non-deliverable forwards" (NDF) markets. For foreign institutional investors who had limited access to the forwards markets on the domestic INR-US$ markets, the NDF market did not suffer from the constraints imposed by capital controls. However, for domestic investors, this led to limited participation by financial institutions of the onshore currency forward market.

Currency options were newcomers in the Indian scenario. The Reserve Bank of India (RBI) launched trading in rupee options from July 7, 2003. The timing could not have been more appropriate, with the government finding itself in a comfortable forex reserves position and the markets being mature enough to exploit the opportunity. On the first day it witnessed brisk activity, trading volumes of over US$200 million, with foreign as well as Indian banks such as Standard Chartered, HSBC, ABN Amro, SBI, IDBI, ICICI Bank and IndusInd entering into major transactions. Large corporate houses such as Reliance, HCL, Murugappa and L&T were not far behind. Before the introduction of currency options, Indian corporations had only two alternatives: either to enter into a forward contract or to leave the exposure open. The problem with forwards was that they were price fixing agreements and denied any gains of favourable movement in the market. Leaving the exposure open subjected the firms to the mercy of the market.

Over-The-Counter (OTC) Currency Derivatives

Over-the-counter or off-exchange trading was the trading of financial instruments directly between two parties instead of going through the exchange. The OTC currency market had traditionally been dominated by orthodox instruments such as outright forwards and FX swaps. The total daily average turnover in foreign exchange OTC markets for the month of April 2007 was US$2.3 trillion as against US$1.3 trillion in April 2004. In comparison, the figure for exchange traded currency contracts was US$72 billion in April 2007 and US$22 billion in April 2004. The percentage share of the Indian rupee

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[3] *Kumar, Ramesh, "How India Inc. Handles Currency Risks," Hindu Business Line, April 12, 2009, http://www.thehindubusinessline.com/bline/iw/2009/04/12/stories/2009041250511200.htm. last accessed on February 15, 2013.*

[4] *Chakrabarti, Rajesh and Sankar De (Eds.), Capital Markets in India, Sage: New Delhi, 2010.*

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in total daily average foreign exchange turnover increased from 0.3 per cent in April 2004 to 0.9 per cent in April 2007.

In order to simplify procedural requirements for the small and medium enterprises (SME) sector, RBI granted flexibility for hedging the underlying exposure as well as future anticipated exposures without going through the rigours of complex documentation formalities. In order to ensure that SMEs understood the risks of these products, only banks with which they had credit relationships were allowed to offer such facilities.

CURRENCY HEDGING

Currency hedging was of prime importance in a commoditized industry like the textile industry, which operated on razor thin margins. In a hyper-competitive globalized environment, the problem of thin margins was compounded when there were adverse currency movements; thus, currency hedging assumed great importance. The story was no different for TT Textiles, where exports formed a major chunk of the company's revenues (see Exhibits 2 and 3 for the financial results of TT Textiles).

TT TEXTILES' HEDGING STRATEGY

In the textile industry, hedging currency risks had traditionally been through forwards, where the company had an estimate of future sales and took a conservative estimate of the US$-INR movement to hedge against unfavourable currency movements. TT Textiles was no different in that it also used forwards to safeguard against adverse currency movements. The options market was not considered attractive by SME players as the option premiums were substantial and the industry margins were quite low, making forwards the most popular of all the instruments used by the company (refer to Appendix 1).

In late 2006, when the Indian rupee was rapidly appreciating and the dollar was depreciating, the Indian textile industry was reeling under severe currency pressures. With the rupee hovering around the 45 INR/US$ mark, bankers were forecasting that the rupee would continue to appreciate — perhaps even to the unprecedented level of INR 35/US$, and the textile industry was looking for solutions in order to stay afloat in those testing times. The appreciation in the rupee was particularly galling as it was driven by foreign portfolio investment flows to India rather than global factors, with the result that other competing or importing countries did not experience a similar currency appreciation and Indian exporters were losing out in competitiveness. For a low-margin industry like textiles, this threatened the very viability of exporters.

There were two basic approaches that companies could take:

- Boost sales in the domestic market, which had started to look increasingly attractive.
- Hedge against the upward movement of the rupee against the dollar. The OTC derivatives market, which was emerging rapidly in India, presented some exciting options at that time.

TT Textiles, which was seeking to protect itself from currency risks, was looking at the derivatives market when an ABC Bank[5] derivatives sales representative paid a visit to the company. TT Textiles, which until then had been much more familiar with forwards as instruments of hedging foreign currency exposure, was introduced to a new instrument called currency swaps. Currency swaps (see Appendix 1) enabled the company to limit its exposure to fluctuations in the rate of the US dollar. The

[5] *Name has been disguised by the author*

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swap was on the US dollar and Swiss francs, as these two currencies traditionally had stable exchange rates in comparison to the rupee and dollar.

THE SWAP DEAL

When the swap deal was designed in 2006, it was based on the fact that, historically, the US dollar had never gone below the exchange rate of 1.09 CHF, making the latter a very stable currency. But Jain was still unsure. He recalled:

> Although I was assured again and again that it has never happened in the history of currency markets that the CHF had fallen to below the 1.09 mark, I wanted to be doubly sure that I was safe. So they offered us a cut-off mark of 1.04 CHF/US$, although our payouts reduced as well.

With TT Textiles keeping the strike rate in the swap deal at 1.04 CHF/US$, it was confident that the likelihood of the strike rate bypassing the set 1.04 mark would be next to impossible.

The notional principal amount of the contract was INR 225 million = US$4965791.22 = CHF 6306554.84.[6] This essentially meant that the interest rate the bank would be paying to TT Textiles would be based on INR 225 million. However, it should be noted that no initial principal exchange took place in this swap. At the end of the swap, TT Textiles had to pay CHF 6306554.84 to ABC Bank and would receive INR 225 million. The contract also implied that TT Textiles would receive a fixed interest of 1.77 per cent semi-annually on the notional amount of INR 225 million and would not pay any interest on the notional amount of CHF 6306554.84 to ABC Bank. This meant that every six months, ABC Bank would pay TT Textiles INR 2 million. In exchange, TT Textiles would have to do next to nothing as long as the exchange rates stayed within reasonable boundaries. ABC Bank was offering TT Textiles a credit limit of INR 80 million on the basis of its balance sheet to cover the margins and mark-to-market losses, so TT Textiles was virtually assured of an earning of INR 12 million over the life of the deal. This would provide a much needed buffer against the squeeze in margins that the appreciating rupee would imply for TT Textiles.

The start date for the swap was October 19, 2006 and the expiration date was October 15, 2009, with a delivery date of October 19, 2009. The expiration date was the last date until which the derivative was valid, and the underlying asset had to be delivered on the delivery date.

The swap also had a *partial barrier* on the CHF/US$ rate of 1.04 within the window of September 15, 2009 to October 15, 2009. A partial barrier meant that if the CHF/US$ rate fell below 1.04, the payouts would be according to the spot rate. In the option, TT Textiles was supposed to receive US$ Put/CHF Call at strike 1.27 for US$4965791.22 with knock-out at 1.04 while ABC bank bought US$ Call/CHF Put at strike 1.27 for US$4965791.22. This implied that TT Textiles had an option to sell US$4965791.22 at 1.27 if the US$/CHF did not trade at or below 1.04 at any time between September 15, 2009 and October 15, 2009. TT Textiles also had an obligation to sell US$ at 1.27 if US$/CHF went above 1.27 on maturity.

Protection Structure on US$/INR

The protection on US$/INR was similar to the US$/CHF, with the only difference being that there was no barrier of any form. TT Textiles had the option of buying US$4965791.22 at 46.25 if the US$/INR exchange rate was above INR 46.25 at maturity. TT Textiles had the obligation to buy US$4965791.22 at INR 45.00 if the US$/INR exchange rate was below 45.00 at maturity. TT Textiles would buy at the spot rate if the US$/INR rate was between INR 45.00 and INR 46.25.

[6] *Exact size of the deal has been modified by the author. Exchange rates implied are 45.31 INR/US$ and 1.27 CHF/US$.*

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Risks

US$/CHF risk: The risk associated with the above deal was the partial barrier, which implied that there was no protection available on CHF notional if US$/CHF traded below 1.04 between September 15, 2009 and October 15, 2009. This basically meant that if the US$/CHF rates touched 1.04, TT Textiles would have to pay INR 41.4 million more than the expected levels. Also, ABC Bank would have an obligation to sell US$ and buy CHF at 1.27 if CHF depreciated beyond 1.27 on maturity.

US$/INR risk: Here the risk was that TT Textiles would lose the upside if the dollar depreciated beyond INR 45.00 and would have an obligation to buy US$ at INR 45.00. Also, if the dollar appreciated against INR beyond INR 45.31, TT Textiles would end up buying US$ at a lower rate. However, the ceiling for this was INR 46.25.

Historically, the CHF/US$ rate had never gone below the level of 1.09, and at the knock-out rate of 1.04, TT Textiles felt confident that they had their bases covered. Their confidence was boosted when other players in the market opted for similar derivative instruments. The deal, it seemed, was simply too good to turn down. TT Textiles entered into an agreement with ABC Bank for a swap contract applicable from September 16, 2006 to October 19, 2009.

The Rupee in 2007

In 2007-08, the net FII inflow into India shot up to US$20.3 billion, which was a key driver for an appreciating rupee, up from INR 45/US$ to INR 39.60/US$ in the year end (see Exhibit 4), and for the overall bullish outlook for the Indian economy as evident from the rising Sensex levels, which went from under 5,000 in early 2004 and peaked just above 21,000 in January 2008. This had a damaging impact on Indian exporters as a rising rupee shrank margins drastically, especially for mid-sized players who did not have economies of scale in production or sophisticated currency risk hedging positions. The woes of the exporters weren't limited to the rise of the rupee against the dollar. Domestic inflation and rising raw material prices further strained their already dwindling profits. For instance, there was a rise in cotton prices globally, which made the procurement of good quality raw material expensive.

A 2008 article in *IndiaKnowledge@Wharton* captured the mood of the time: "Our competitiveness for the time being has gone away," said P.D. Patodia, chairman of the Confederation of Indian Textile Industry. The association estimates that for every 1 per cent fall in the value of the dollar compared with the rupee, profit falls by 1.2 per cent. "Some exporters will be permanently damaged and not all will survive," said Subir Gokarn, chief economist for Standard & Poor's Asia-Pacific.[7]

The Tide Turns in 2008

The rupee, however, swung completely in the opposite direction the following year when the financial meltdown led to massive FII outflows from the emerging market economies. FII outflows in India exceeded US$11.1 billion during the first nine and a half months of calendar year 2008, of which US$8.3 billion occurred over the first six and a half months of the financial year 2008-09. This led to severe rupee depreciation in the year 2008 with the rupee touching INR 50 per dollar. At the same time, the stock market collapsed and the Sensex crashed from close to 21,000 in January to just over 8,600 in November 2008.

During the course of the year, the recession in major textile importing economies such as the United States, European Union and Japan adversely affected textile exports from India. Anti-dumping,

[7] *"A Stronger Rupee Tests the Fabric of India's Textile Exporters," India Knowledge@Wharton, February 7, 2008, http://knowledge.wharton.upenn.edu/india/article.cfm?articleid=4258 last accessed on February 15, 2013.*

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safeguard duty and other protectionist measures adopted by various countries proved to be major stumbling blocks for exports. Moreover, the government of India did not step up its export subsidies, unlike China, Vietnam and Bangladesh, where subsidies were increased to 16 per cent, 15 per cent and 14 per cent respectively, giving an edge to the exporters there. The growth rate in the textile industry fell to 0.8 per cent in 2008-09 (April-August). Domestic demand was also affected on account of marked inflation and a fall in GDP growth. The Indian government also withdrew export incentives at the first signs of rupee depreciation, and only partially restored them later. It was indeed a testing time for Indian textile manufacturers.

The Rollercoaster Ride

Everything was moving along in the expected manner for TT Textiles when it received the first cheque of INR 20 lakh (2 million) from ABC Bank. As the rupee started falling, the belief that the rupee would get stronger (if only to get back to where it was) grew firmer. TT Textiles started increasing its exposure to this instrument. Other firms were also making extra money out of this hedging strategy. In a 2008 *Business Standard* article, Ranju Sarkar described the scenario at the time:

> "It all began four-five years ago, when companies began taking small exposures of Rs 1-2 crore in currency swaps, options. As they began making money, they started increasing their exposures. We made profits and entered into more contracts … ," said a CEO.
>
> What's interesting is that it's not just exporters or companies with foreign currency exposures that went for derivative deals. Companies have swapped their rupee loans to Swiss Franc to get an interest benefit of 2-2.5 per cent, due to the carry that was available, and the risk they were willing to take.
>
> "It seemed like a reasonable risk to take if the franc hasn't breached 1.10 level against the US dollar for 20-25 years," said a forex consultant.
>
> "Companies were not required to make any upfront payment and there was no risk, so companies went for it … ," explained the forex consultant.
>
> A majority of the deals were in Swiss Franc and were executed at a rate of 1.25 franc for every dollar. Banks and companies thought they were protected against the movement of the franc against the dollar as the lifetime lowest rate was 1.10; people thought it's highly unlikely that the franc would breach these levels.
>
> The options with knock-ins meant that if the franc breaches a level below this, say 1.09 to the dollar, then the company has to pay the difference.
>
> "The deals were executed when the franc was trading at 1.25 to the dollar while today it is trading at 1. They sold at 1.25 and now have to buy at the rate of 1, at a 25 per cent loss," explained the CFO of a leading corporation in Ludhiana.
>
> Nobody thought of this kind of forex movement. Earlier, an option would get knocked out within a week (if a company had entered into a contract at 1.25, in one week the option got knocked out at, say 1.30). The company would do another deal at 1.20, make money, and do another deal at 1.15," said the CFO.[8]

By the end of 2007, the United States had started showing signs of recession. But the belief that the United States was decoupled from India was strong among Indian corporations and investors,

[8] *Sarkar, Ranju, "Forex Derivative Losses Haunt Ludhiana Firms," Business Standard, April 11, 2008, http://www.rediff.com/cms/print.jsp?docpath=//money/2008/apr/11forex.htm last accessed on February 15, 2013.*

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which was reflected in the inflating Indian stock market. In January 2008, a rude shock came in the form of a major crash in the Indian stock market when the Sensex showed decrements to the order of 2,000 points. At that time, the US$/INR exchange rate was close to 39.60 INR/US$. The stock market crash led to panic among foreign investors and the FDI outflows from the Indian economy grew, which made the exchange rate creep up to 41 INR/US$ and eventually back to the 50 INR/US$ mark (see Exhibit 5 for the exchange rate movement of INR-US$). During this time, as Jain jocularly points out:

> Firms started selling when the rupee was at 39 INR/US$ and then at 41 INR/US$ and so on and so forth because the economic forecast was nervous enough to predict the range within a couple of rupees or so. After 46 INR/US$, they gave up selling and when it finally stabilized at 50 INR/US$, no one was selling because by then the prediction had turned to 55 INR/US$! And thus the corporations went on over-hedging their futures.

As many currency derivative deals soured, there were also allegations of misselling by banks. Sarkar wrote:

> A senior executive with a Ludhiana-based company said almost every large company or exporter has dabbled in them. … Experts (forex consultants and CFOs) estimate the notional losses on derivative products in Ludhiana to be Rs 200 crore to Rs 300 crore (Rs 3 billion), with a prominent textile player leading the table. But no company is willing to talk about their exposure or losses. … Companies are obviously circumspect about giving out figures on how much mark-to-market losses they are carrying. But losses are pretty high, spread across 20 to 25 big and small companies, said a forex expert, who tracks the companies in Ludhiana.
>
> Two textile companies have gone for litigation. Nahar Industrial Enterprises, part of the Rs 1941-crore (Rs 19.41 billion) Jawaharlal Oswal Group (Nahar) has moved court against Axis Bank while Garg Acrylics Ltd, part of the Rs 600-crore (Rs 6 billion) Garg Group with interests in steel and textiles, has moved court against ICICI Bank.[9]

The Swiss Franc

Meanwhile, in Europe, the situation was completely different as the European economy had not yet showed signs of recession, and investments in Europe were still considered relatively safe. The same could not be said of the United States. As a result, the CHF started becoming stronger than the US$ and the CHF/US$ rate — which had never breached the 1.09 mark — started decreasing and eventually crossed the strike rate of 1.04 that was specified in the swap deal of TT Textiles, and continued its downward trend to fall below 1. The projected losses for TT Textiles if the CHF/US$ rate touched 0.93 were estimated at around INR 45 million, and with further decrement in the dollar rate, the losses were expected to cross INR 55-63 million (see Exhibit 6 for the exchange rate movement of CHF-US$).

At the same time, the overall economic scenario presented liquidity challenges to the banking sector and forced it to invoke margin calls on the counterparties. TT Textiles was also under tremendous pressure as sales were hit hard, and the company demanded a credit limit increase over and above the INR 4.5 crore credit limit already granted by ABC Bank on the swap deal. However, ABC Bank did not entertain this request, and on the contrary, argued for the removal of the existing credit limit as well. The resultant impact could be seen in the respective year's director's report (see Exhibit 4).

[9] *Sarkar, 2008, Ibid.*

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The Crisis Comes to a Head

Since ABC Bank had entered into similar agreements with a host of other counterparties, there was increasing pressure on the derivatives desk to recover the money from them. The situation was further aggravated by the fact that most of the corporate players had entered the deal thinking it would be an extremely safe bet. They had consistently benefitted from the deal until that point, and these sudden and unforeseen events came as a rude shock to them. Considering the circumstances, bankers from ABC offered a few suggestions. The first suggestion was that TT Textiles should enter option contracts to seek protection; however, the available option contracts were extremely high-priced at that time due to the unstable economic environment. The other suggestion was to cut down on existing positions to reduce the existing exposure of TT Textiles.

Jain sought opinions from a host of financial experts about their expectations from the CHF/US$ exchange rate, but he was not able to form a clear view as there were diverse opinions among the experts due to a lack of clarity on the underlying reasons behind the crisis. The CHF continued to gain strength over the US dollar and touched the 0.96 mark. At this point, the mounting panic even led to speculation that this rate could well reach the 0.90 level. Some firms decided to cut their losses to the extent of 50 per cent of their initial positions in the instrument. The knock-out option in the contract was dependent on the event that the exchange rate touched the 1.04 mark during the *last one month* of the deal.

THE SHORTING OF CHF

The dollar started reversing from 0.96 CHF/$ and gained momentum to reach the 1.17 CHF/$ level in a matter of months (see Exhibit 6). At this point, Jain found himself in an interesting situation. The position still showed a mark-to-market loss, which made the decision to square it off difficult. Circumstances indicated that he should hold the security till maturity since the general belief was that the rate would not slide back to the 1.04-1.05 CHF/$ levels again in the near future. He decided in favour of carrying out a reverse transaction when the CHF/US$ exchange rate was 1.17 CHF/$. This meant that if the rate fell below 1.17 CHF/$, TT Textiles would cover the existing position at the lower rate, say 1.12 CHF/$. These range-bound deals made some money for TT Textiles during that period.

While the rates stood at 1.17 CHF/US$, the special month of the deal, i.e., September 2009, was still nearly six months away. The instrument carried a clause that made TT Textiles liable to losses only if the CHF/US$ exchange rate touched the 1.04 CHF/US$ mark during the special month.

With recession hitting the United States and the subsequent weakening of the dollar in 2008-09, the CHF/US$ exchange rate had sunk to the unprecedented low of 0.96 CHF/$. Once it bounced back, however, it put TT Textiles and Jain in a Catch-22 situation. He had two choices: He could either continue with the swap deal of CHF/US$, which was due to expire in October 2009; or he could exit the swap deal at this favourable juncture and put an end to any uncertainty. By getting out of the position, he might, on the one hand, forego attractive returns on the swap if the dollar remained at a high rate, but on the other, he would be protected against any adverse movement of the dollar.

Time to Decide

The sun had shifted but the furrows on Jain's forehead lingered. Having witnessed the unexpected and alarming behaviour of the supposedly steady exchange rate relationship between the US dollar and the Swiss franc, he was still undecided about the right course of action. Should he close out his positions, taking a hit to the order of INR 9-19 million or should he trust that the US dollar would be able to maintain its current position and thus run the risk of exposing his firm to losses to the tune of INR 55-63 million (figures masked). Time was running out and a critical decision had to be made.

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APPENDIX

Forwards: A contracted agreement specifying an amount of currency to be delivered at an exchange rate decided on the date of contract.

Swap: Traditionally, the exchange of one security for another to change the maturity (bonds), quality of issues (stocks or bonds), or because investment objectives have changed. Recently, swaps have grown to include currency swaps and interest rate swaps.

No margin money required on the part of TT Textiles as ABC Bank approved a credit limit of INR 50 million based on the balance sheet of the firm.

Currency option: A contract that gives the owner the right, but not the obligation, to take (call option) or deliver (put option) a specified amount of currency at an exchange rate decided at the date of purchase

Option legs: A type of order that allows an option trader to simultaneously buy or sell a number of different options that traditionally could only be achieved by placing separate orders. This type of order is primarily used in multi-legged strategies such as a straddle, strangle, ratio spread and butterfly.

Knock-out Option: An option with a built in mechanism to expire worthless should a specified price level be exceeded. This is an exotic option mainly used for commodities and currencies.

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EXHIBIT 1

SNAPSHOT OF WORLDWIDE OPERATIONS OF TT TEXTILE



Source: Company website

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EXHIBIT 2

FINANCIAL RESULTS - YEAR 2009 (INR in LAKHS)

S. No.	PARTICULARS	31.03.2009
1	a) Net Sales/ Income from Operations	26744.15
	b) Other Operating Income	182.55
2	Expenditure	
	a) (Increase) Decrease in Stock in trade	–227.71
	b) Consumption of raw materials/	19586.69
	Purchase of goods for resale	
	c) Staff Cost	703.28
	d) Depreciation	777.37
	e) Other Expenditure	5779.65
3	Profit /(Loss) from operations before Int & tax	307.42
4	Other Income	188.54
5	Profit /(Loss) before interest	495.96
6	Interest	1786.77
7	Profit /(Loss) from Ordinary Activities Before Tax	–1290.81
8	Tax Expenses	
	- Income Tax(Current)	--
	- Fringe Benefit Tax	4.54
	- Deferred Tax	–295.98
	Adjustment foe MAT credit entitlement	--
9	Net Profit /(Loss) from Ordinary Activities After Tax	–999.37
10	Paid-up Equity Share Capital	2149.8
	(Face value per share in INR)	10
11	Reserve Excluding Revaluation Reserve	1777.39
12	Basic & Diluted EPS (in INR) not annualized	–6.00
13	Public Shareholding	
	- Number of Shares	10273115
	- Percentage of Shareholding	47.79%
14	Promoters and Promoters' Group Shareholding	
	a) Pledged/ Encumbered	
	- Number of Shares	NIL
	- Percentage of Shares (as a % of shareholding of promoter and promoter group)	NIL
	- Percentage of Shares (as a % of shareholding of the company)	
	b) Non-encumbered	NIL
	- Number of Shares	11224935
	- Percentage of Shares (as a % of shareholding of promoter and promoter group)	100%
	- Percentage of Shares (as a % of shareholding of the company)	52.21%

Source: From Company reports (Modified by the author)

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EXHIBIT 3

FINANCIAL RESULTS

SNo.	Particulars	Quarter Ended 30/9/2008	Corresponding Quarter Ended 30/9/2007	Half Year Ended 30/9/2008	Corresponding Half Year Ended 30/9/2007	Previous Year Ended 31/3/2008 (Audited)
1	(a) Net Sales/Income from Operations	5931.90	6405.96	13982.77	12416.69	65033.98
	(b) Other Operating Income	23.52	23.02	53.72	32.40	21.14
2	Expenditure					
	a) (Increase) Decrease in Stock in trade	(40.54)	628.54	(428.37)	789.95	30.05
	b) Consumption of raw materials/ Purchase of goods for resale	4433.85	3931.23	11390.55	8459.92	55238.57
	c) Staff Cost	138.30	128.72	303.44	253.09	696.90
	d) Depreciation	197.36	157.63	358.11	301.38	643.11
	e) Other Expenditure	857.02	986.05	1740.78	1684.58	6837.84
3	Profit from Operations before Inn, exceptional Items &Tax	369.43	596.81	671.98	960.17	1608.65
4	Other Income	118.01	0.00	267.08	0.00	257.09
5	Profit before Interest & Exceptional Items	487.44	596.81	939.06	960.17	1865.74
6	Interest	448.11	537.98	837.24	772.11	1531.96
7	Profit after interest but before exceptional items	39.33	58.83	101.82	188.06	333.78
8	Exceptional Items	0.00	0.00	0.00	0.00	0.00
9	Profit from Ordinary Activities Before Tax	39.33	58.83	101.82	188.06	333.78
10	Tax Expenses					
	- Income Tax (Current)	4.85	6.66	11.53	21.30	37.03
	- Income Tax (Earlier Years)	-			-	1.03
	- Fringe Benefit Tax	1.71	1.22	3.49	3.40	6.92
	- Deferred Tax	-			-	11.43
	Adjustment for MAT credit entitlement	(4.85)	(6.66)	(11.53)	(21.30)	(37.03)
11	Net Profit from Ordinary Activities After Tax	37.62	57.61	98.33	184.66	314.40
12	Paid-up Equity Share Capital	2149.80	2149.80	2149.80	2149.80	2149.80
	(Face value per share in INR)	(10.00)	(10.00)	(10.00)	(10.00)	(10.00)
13	Reserve excluding Revaluation Reserve		-		-	3223.36
14	Basic &Diluted EPS (in Rs.) not annualised	0.18	0.27	0.46	0.86	1.46
15	Public Shareholding					
	- Number of shares	10339361	10339613	10339361	10339613	10339361
	- Percentage of Shareholding	48.09%	48.10%	48.09%	48.10%	48.09%

Source: From Company reports (Modified by the author)

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EXHIBIT 4

EXCERPT FROM DIRECTOR'S REPORT (2008-09)

REVIEW OF OPERATIONS AND FUTURE OUTLOOK

The year 2008-09 was the most challenging year in the history of the Company. It has also been acknowledged to be the worst year for the world economy since the great depression of 1932 in the United States of America. The current global recession is much more widespread than the great depression of the last century. Economists, Central Banks, Finance Ministers and Heads of States have not been able to figure out the exact cause of the global meltdown and the deep recession. Economists are still not able to say when the revival of the economy will start.

Several large banks in the USA, England, Canada and in many other developed and developing countries have failed. Many large corporates and MNCs have gone bankrupt. Export markets saw massive shrinkages. Your Company due to its engagement in the most severally impacted sectors like exports and textiles could not escape unhurt and for the first time since its inception incurred losses amounting to Rs. 36.00 Cr. The need for maximum stimulus to export and textile sectors for their revival has been recognized by the Textile Ministry and other Government Agencies.

About one third of the Company's losses arose mainly due to forward sale of foreign exchange in anticipation of cotton exports during the period from November 2008 to February 2009. Following the general trend and analysts' view during the first quarter of 2008 when the dollar was at Rs.40.00, your Company had sold dollars, roughly equivalent to 2 months cotton exports. A steep depreciation in the value of the rupee vis-a-vis the US dollar by about Rs.10 to 12 per US dollar during the delivery period eventually resulted in a loss to the Company to the extent of the exchange difference. In addition, as a result of substantial increase in the Minimum Support Price (MSP) of Kapas up to 40 per cent by the government during the recession period, your Company found export of Raw Cotton unviable and had to suffer Foreign Exchange losses. A strengthening Interest rate regime during the year under review also contributed to higher finance charges.

Other factors that exacerbated the losses were a significant fall in cotton and other commodity prices because of a global meltdown and consequent write-down in inventory values by 20 per cent to 30 per cent, increase in power cost and non availability of power in Tamil Nadu and withdrawal of certain export benefits during the relevant period.

The Government of India arranged Kapas procurement through its agencies like CCI, NAFED, Maharashtra Cotton Federation etc. These agencies purchased almost 60 per cent of the Indian crop at the minimum support price and in the process recorded loss of up to Rs. 4000 per candy i.e., approximately 20 per cent of the price. In view of the non-viability of the cotton procurement your Company had no choice but to almost totally suspend cotton ginning and oil seed crushing operations and also shelve cotton trading operations during the year under review.

Your Company has also lost a total turnover of Rs. 400 Cr comprising Rs. 350 Cr of cotton exports and Rs. 50 Cr of domestic sale of cotton. It was not possible to substitute this turnover by any other profitable alternatives due to prevalent turbulent market conditions and adverse export prices.

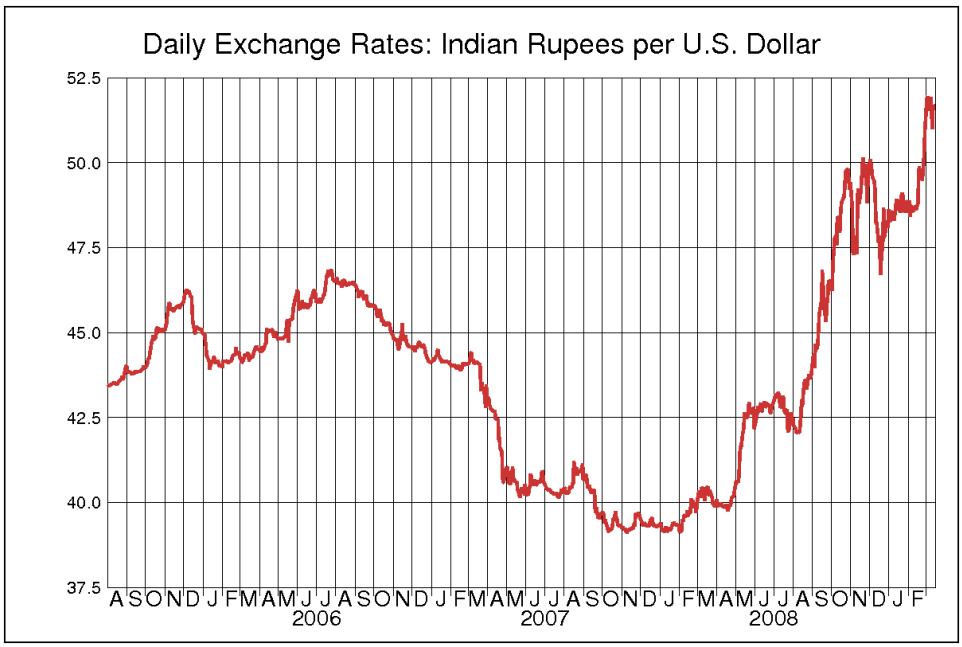
Source: Company reports

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EXHIBIT 5

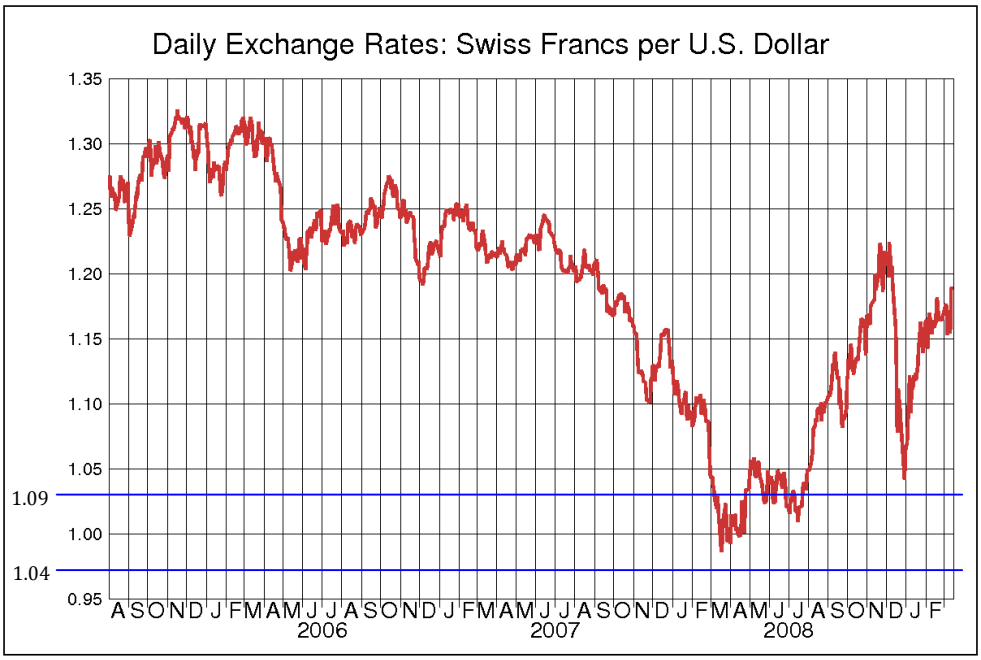
EXCHANGE RATE MOVEMENT (2004-2009) OF US$-INR CURRENCY PAIR



Source: http://fx.sauder.ubc.ca/ last accessed on February 15, 2013.

EXHIBIT 6

EXCHANGE RATE MOVEMENT (2004-2009) OF US$-CHF CURRENCY PAIR



Source: http://fx.sauder.ubc.ca/ last accessed on February 15, 2013.

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